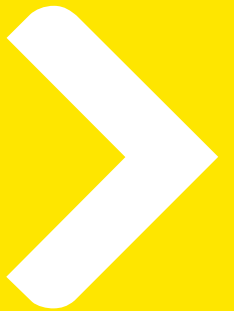




ITEXA - 2026-02-11 - Seminar 01

Application development lifecycle: pitfalls to avoid

Miroslav Jasso | Head of IT



Presentation Contents

- 01** Introduction of RPC & lecture structure
- 02** Overview of the Application Development Lifecycle
- 03** Stage 1 - Planning
- 04** Stage 2 - Analysis
- 05** Stage 3 - Design
- 06** Stage 4 - Implementation
- 07** Stage 5 - Testing
- 08** Stage 6 - Deployment and Maintenance
- 09** Your Feedback



Introduction

Miroslav Jaššo, Head of IT; Simona Černáková, Executive Assistant



Miroslav Jasso ✓

Senior Director | Information Technology | IT Operations & Innovation | Digital Transformation & Program Management | Information Security & Strategic Communication | DORA & IT Resilience | 25+ years IT Experience



Head of Information Technology

RPC • Full-time

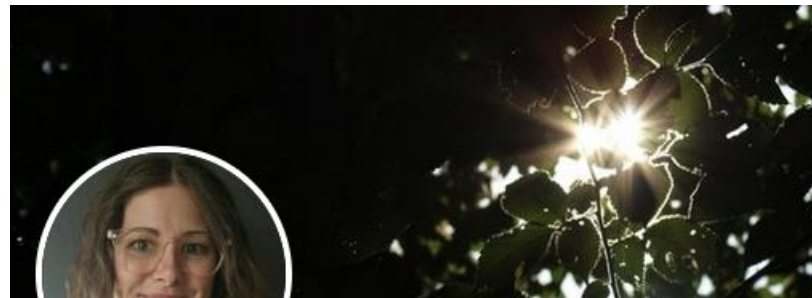
Oct 2022 - Present • 3 yrs 5 mos

Bratislava, Slovakia • Hybrid

Contributing to growth of processed transactions from 1.5 billion in 2022 to 2.5 billion in 2024 and sponsoring technical readiness to enter 3rd party market in 2023.

Leading a 120+ strong IT team consisting of IT Architecture, IT Infrastructure, IT Operations, Development, Quality Assurance, Delivery Data Services, Delivery Online Payment Services, Data Warehouse, Service Configuration and Authorization and taking responsibility for 40%+ of the company's workforce.

Managing yearly IT budget of 10+ mil EUR in both OPEX and CAPEX.



Simona (Gregušíková) Černáková • 1st

Executive Assistant to GM at Regional Card Processing Centre

Slovakia • [Contact info](#)

[207 connections](#)

Experience



Regional Card Processing Centre

Full-time • 10 yrs 9 mos

- **Executive Assistant to GM**
Jun 2016 - Present • 9 yrs 9 mos
- **Service Desk Specialist**
Jun 2015 - Jun 2016 • 1 yr 1 mo

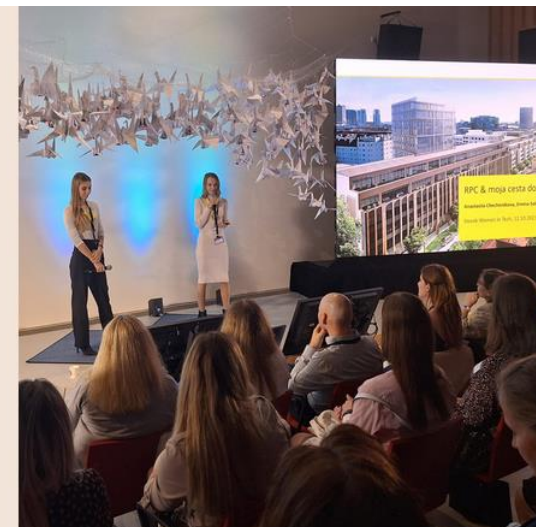


Leading provider of innovative and reliable card services

As part of Raiffeisen Bank International, Raiffeisen Processing Centre provides secure and reliable card services to Raiffeisen banks and other customers and develops innovative payment solutions

We motivate future generations to consider IT careers

We empower young generations to see STEM (science, technology, engineering, and mathematics) as an interesting direction for their future. In 2023, we started cooperation with Aj ty v IT, where we support projects inspiring girls and women to start their career in IT – Girl's Day and Slovak Women in Tech. We also launched a new program ITEXA for students of the University of Economics in 2024 to provide real insights, use-cases and information from our experienced professionals.



13
countries

2.5
billion
transactions

260+
colleagues

8 300
000
cards



--- Section1: Basic Information ---

- **Duration:** 12 lectures, each 90 minutes. It is possible to request a double-lecture – 180 min duration for more complex topics
- **Start / End:** 18.2 – 8.5.2026 (every Wednesday 9:15 – 10:45)
- **Lecturing:** on-site at the faculty or on-site in RPC
- **Themes for lectures:** in next section, including exact timing
- **Grades:** students will be graded based on performance during lectures and quality of delivered projects (details in next section)
- **Attendees:** cc. 20 students
- **Academic Garant:** Associate Professor SCHMIDT, Peter, Ing. Mgr., PhD.
- **Sponsor/Coordinator on RPC side:** Miroslav Jasso / Simona Cernakova

--- Section2: Themes for lectures, dates & time, lecturers ---

	Date of lecture	Site	Timing	Name of lecture	Department	Lecturer
1	11.2.2026	FHI EU	90 min	Introduction + Application development lifecycle – pitfalls to avoid	IT	Miroslav Jasso
2	25.2.2026	FHI EU	120 min	Introduction to Business Analysis	B&I	Rastislav Vallo Roman Klein
3	4.3.2026	RPC	120 min	"Importance of (card) payments in the banking industry of today and tomorrow." – 30 min Enterprise Architecture – Software- 90 min	IT	Marko Rankovic Michal Bielik (Martin Sedlacek)
4	11.3.2026	FHI EU	90 min	IT Infrastructure Architecture – how to make your life easier	IT	Marek Mintal (Miroslav Jasso)
5	18.3.2026	FHI EU	90 min	Design of data model, ORACLE, POSTGRE SQL, DB management, IT analysis – how to ask proper questions	IT	Michal Vadovic (Miroslav Jasso)
6	25.3.2026	FHI EU	90 min	+ REST API architecture	IT	Jakub Rapant (Miroslav Jasso)
7	1.4.2026	FHI EU	90 min	UX and front-end development	IT	Martin Mozi (Michal Bielik)
8	8.4.2026	FHI EU	90 min	GITLAB system, version control, code co-development and managed code deployment, Frontend development – Single page applications	IT	Martin Mozi (Michal Bielik)
9	15.4.2026	RPC	90 min	Efficient Software Integration and CI/CD Deployment	IT	Martin Mozi (Michal Bielik)
10	22.4.2026	RPC	180 min	Testing and test plan	IT	Dusan Ivanoc, Martin Simon, Kristina Dienes, Jozef Slachta
11	29.4.2026	FHI EU	90 min	Handover Application to Operation/Daily run. Operation duties and ITSM (Incident, Problem, Change and Release Management)	O&T	Miroslav Strecok Barbora Kovacova
12	6.5.2026	FHI EU	90 min	Security risks from the perspective of DevSecOps	SEC	Branislav Jancovic (Matej Koszoru)

--- Section3: Students evaluation & semester project ---

Predmet „Programovanie webových a mobilných aplikácií II“ bude hodnotený na základe aktivity na prednáškach (40%) a vypracovaní záverečnej práce (60%). **Celkový počet bodov je 20. Záverečná práca bude hodnotená na základe splnenia zadania a kvality riešenia 0-12 bodmi. Za aktivitu na prednáškach sa dá získať 0-8 bodov.** Pre úspešné hodnotenie predmetu je nutné získať minimálne 50% bodov z oboch častí (čiže min. 4 body za aktivitu na prednáškach a min. 6 bodov za záverečnú prácu)

Výsledná známka:

A (18-20 bodov)

B (16-17 bodov)

C (14-15 bodov)

D (12-13 bodov)

E (10-11 bodov)

Fx (menej ako 10 bodov)

Téma záverečnej práce – vytvorenie webovej aplikácie v skupinách po 4-5 študentov.

1. Križovatka
2. Kyvadlo
3. Pružina
4. Šikmý vrh

Očakávané termíny:

- Architektúrna časť do 1.4.2026
- Vývoj/development do 22.4.2026
- kompletné zadanie do konca semestra



Zadanie projektu: Simulátor matematického kyvadla

Úvod

Vytvorte webovú aplikáciu pre real-time simuláciu matematického (rovnoramenného) kyvadla. Aplikácia demonštruje rozdiely medzi analytickým riešením (platné len pre malé uhly) a numerickými metódami (potrebné pre veľké výchylky). Backend v Pythone zabezpečuje fyzikálne výpočty, frontend v JavaScripte vizualizáciu a WebSocket jednosmernú komunikáciu v reálnom čase.

Kľúčové charakteristiky:

- Oddelenie konfigurácie fyzikálneho systému a spustenia simulácie
- Demonštrácia rozdielov medzi malými a veľkými osciláciami
- Konfigurácie obsahujú fyzikálne parametre a numerickú metódu
- Pre malé uhly ($\theta_0 < 5^\circ$): analytické riešenie je presné
- Pre väčšie uhly: nutné numerické metódy (nelineárna rovnica)
- Rovnakú konfiguráciu možno spustiť s rôznym trvaním
- REST API slúži na správu konfigurácií a riadenie simulácií
- WebSocket je jednosmerný (server → klient) - prenos simulačných dát

Architektúra komunikácie

Communication Flow:

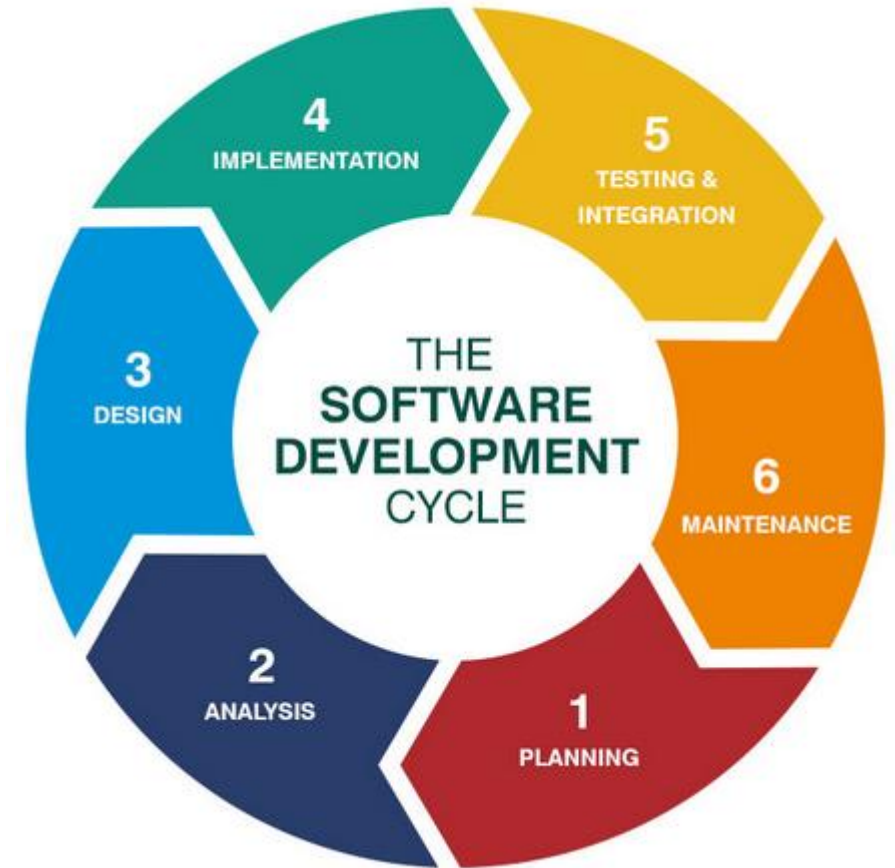
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FÁZA 1: VYTVORENIE KONFIGURÁCIE
1. Používateľ vyplní formulár (dĺžka, hmotnosť, tlmenie, počiatočný uhol, krok, metóda)
   ↓
2. Frontend → REST API: POST /api/pendulum/configurations
   ↓
3. Backend: Validácia parametrov → Výpočet charakteristik → Klasifikácia typu oscilácií
   ↓
4. Backend → Frontend: JSON response (config_id, charakteristiky, varovanie o platnosti analytického riešenia)
   ↓
5. Frontend: Zobrazenie uloženej konfigurácie v zozname

FÁZA 2: SPRÁVA KONFIGURÁCIÍ
6. Frontend → REST API: GET /api/pendulum/configurations
   ↓
7. Backend → Frontend: Zoznam všetkých uložených konfigurácií
   ↓
8. Frontend: Zobrazenie zoznamu s možnosťou výberu

FÁZA 3: SPUSTENIE SIMULÁCIE
9. Používateľ vyberie konfiguráciu a nastaví trvanie simulácie
   ↓
10. Frontend → REST API: POST /api/pendulum/simulations/start
    Body: {"config_id": "conf_abc123", "duration": 20.0}
    ↓
11. Backend: Načítanie konfigurácie → Vytvorenie simulácie → Spustenie na pozadí
    ↓
12. Backend → Frontend: JSON response (simulation_id, websocket_url)
    ↓
13. Frontend: Otvorenie WebSocket spojenia (ws://localhost:8000/ws/{simulation_id})
    ↓
14. Backend → Frontend: Prvá správa (type: "setup")
    ↓
15. Backend → Frontend: Pravidelné správy (type: "data") každých 50ms
    ↓
16. Backend → Frontend: Posledná správa (type: "completed")
    ↓
17. Frontend: Zatvorenie WebSocket spojenia
```




- **Application Development Lifecycle is a framework that guides the development process from idea to deployment and beyond.**
- Often, the Application Development Lifecycle can be **iterative** and often involves **revisiting** certain stages.
- Application Development Lifecycle consists of:
 - Stage 1 - Planning
 - Stage 2 - Analysis
 - Stage 3 - Design
 - Stage 4 - Implementation
 - Stage 5 - Testing
 - Stage 6 - Deployment and Maintenance



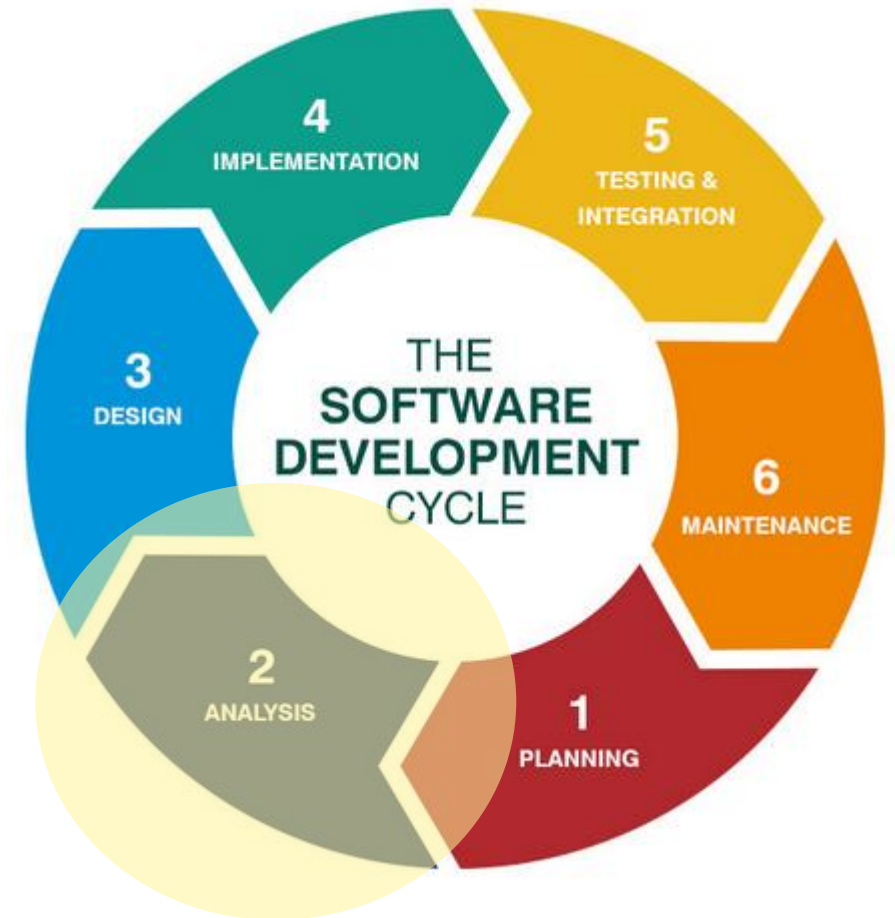


- **Objective:** Emphasizing the importance of thorough planning.
- **Details:**
 - Market research: Analyzing competitors, understanding market needs.
 - Target audience: Identifying user demographics, preferences, and behaviors.
 - Goal setting: SMART goals for what the application aims to achieve.
 - Resource allocation: Budget, time, personnel, and technology requirements.
 - ***AI: data-driven visuals, competitor comparison, goals setting.***



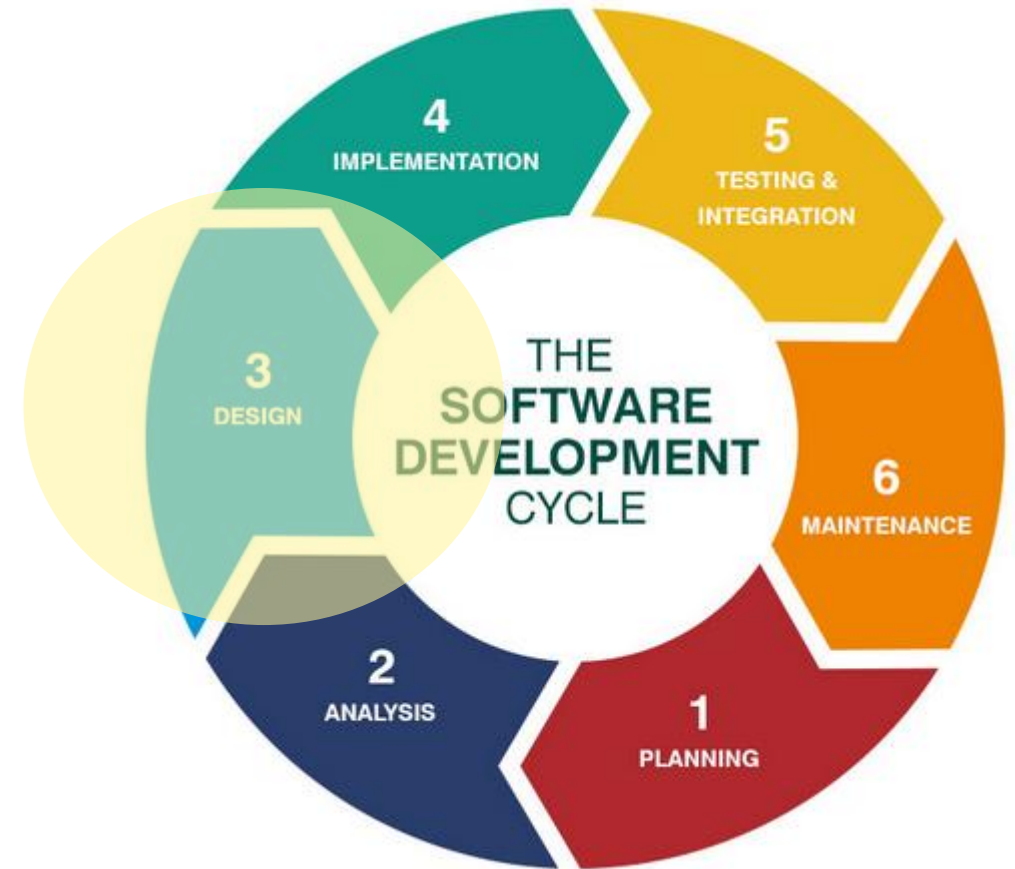


- **Objective: Detailing the process of requirements gathering and analysis.**
- **Details:**
 - Distinguishing between functional (what the app will do) and non-functional requirements (how the app will perform).
 - Developing use cases and user stories to model interactions and desired outcomes.
 - Tools like UML diagrams for system architecture and flowcharts for process visualization.
 - ***AI: extracting and cataloging requirements from stakeholder documents, use cases & user stories generation.***



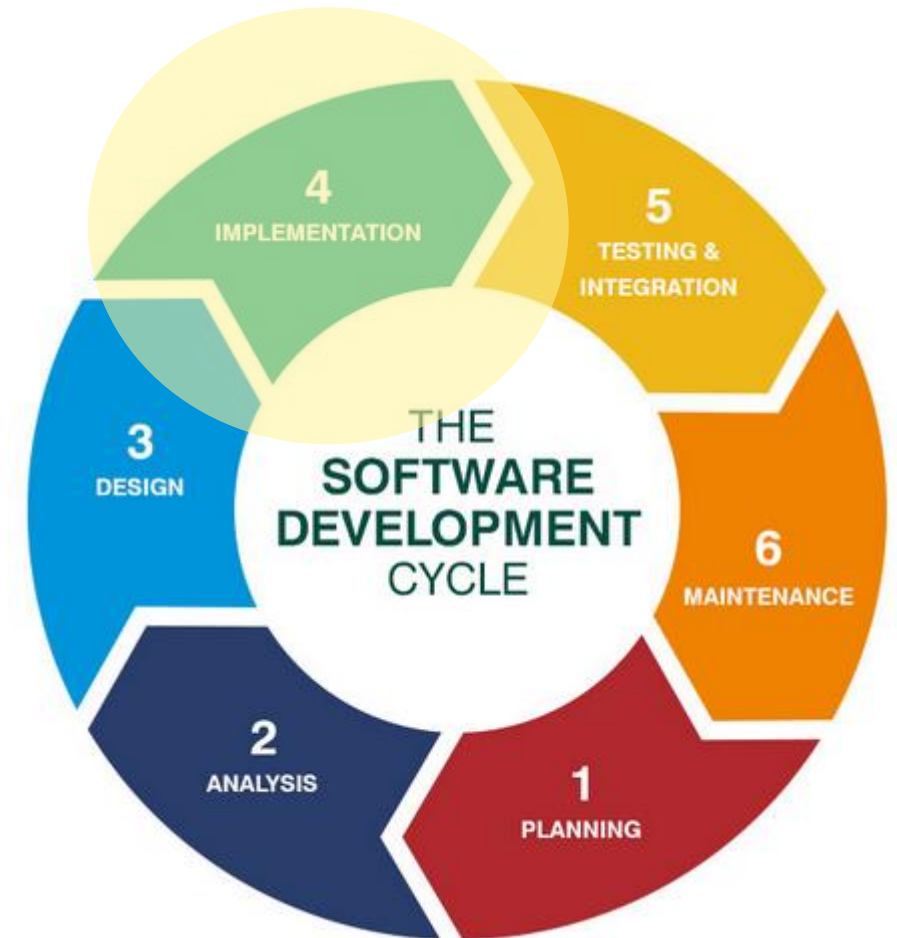


- **Objective: Highlighting the critical role of design in user experience and application functionality.**
- **Details:**
 - UX/UI Design: Principles of designing for engagement and usability.
 - Prototyping: Creating mockups and prototypes to visualize the app.
 - Accessibility: Ensuring the app is usable by people with various disabilities.
 - Responsive design: Designing for multiple screen sizes and devices.
 - ***AI: UI/UX improvements based on behavior analytics, Automated prototyping tool generating mockups.***



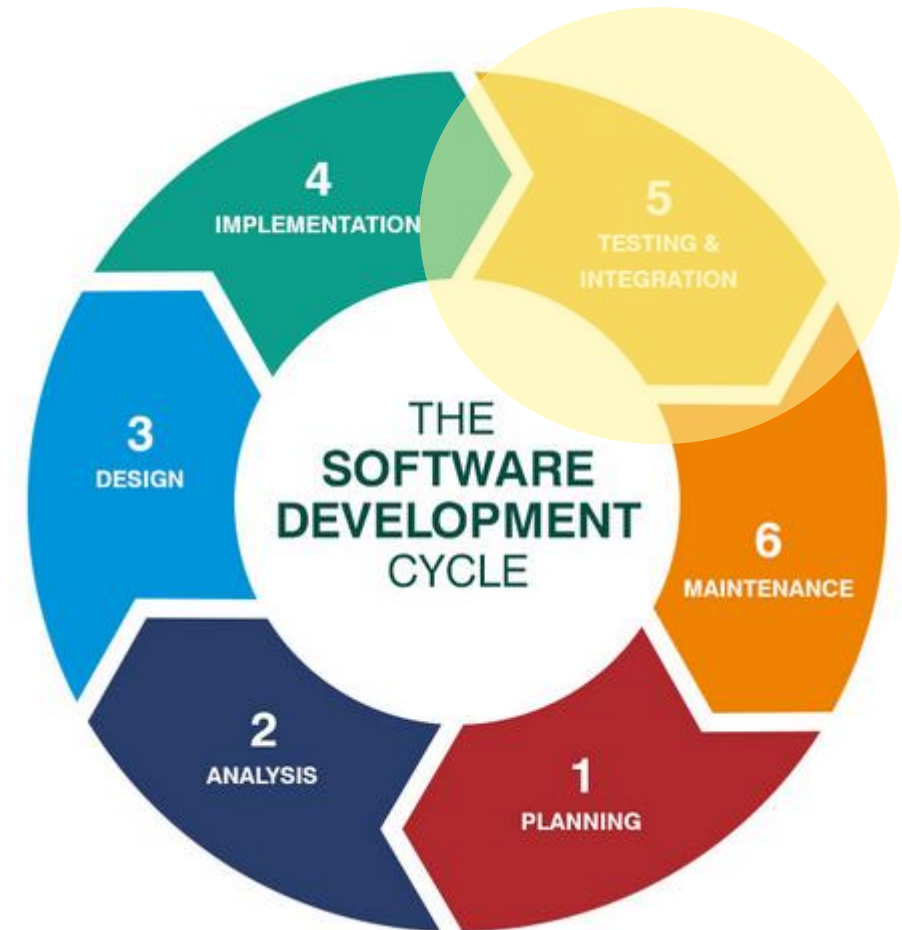


- **Objective:** Discussing the development phase, focusing on coding practices and tools.
- **Details:**
 - Coding standards and practices that lead to clean, maintainable code.
 - Importance of version control (e.g., Git) for team collaboration.
 - Introduction to CI/CD for automating testing and deployment processes.
 - Strategies for effective code review and programming best practices.
 - ***AI: code writing assistants, code review, enforcing coding standards, bug detection, prediction of build failures.***





- **Objective:** Explaining the different types of testing and their importance.
- **Details:**
 - Breakdown of testing types with examples: Unit testing for individual components, integration testing for combined parts, system testing for end-to-end functionality, and acceptance testing for user requirements.
 - Automated vs. manual testing: When to use each.
 - Tools for testing: Overview of popular testing frameworks and bug tracking systems.
 - ***AI: automated test cases scenarios generation and execution, known bugs detection.***

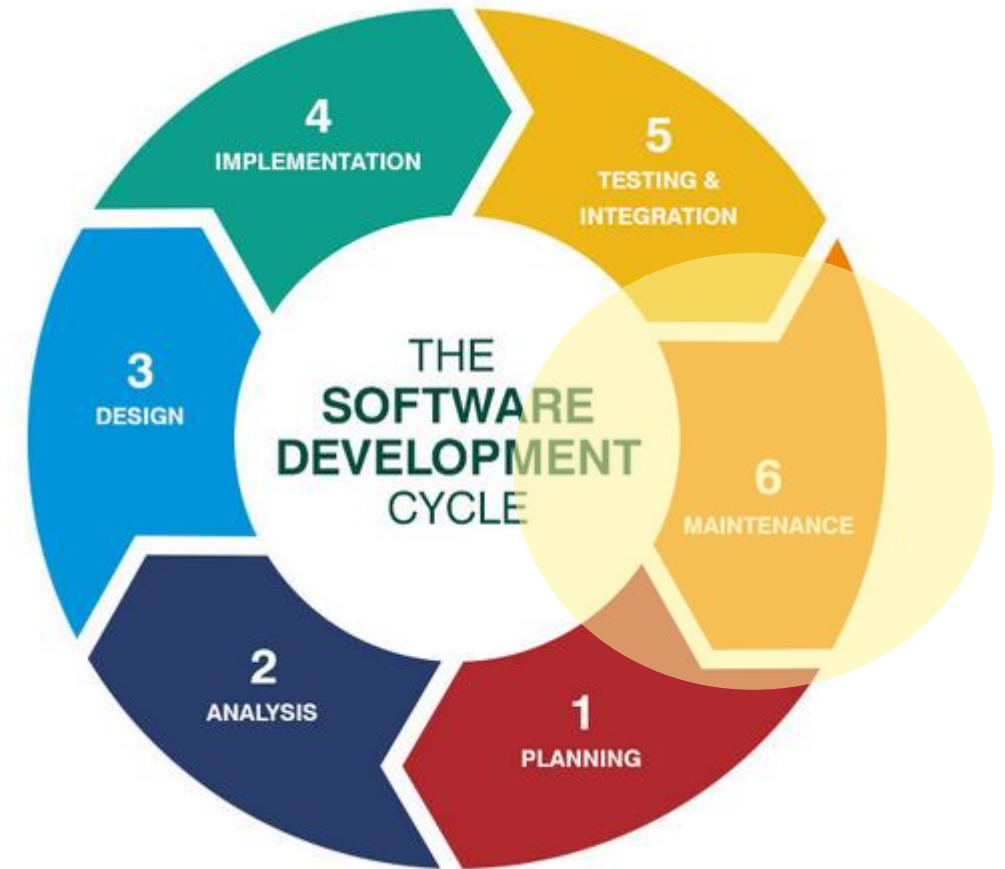




Overview of the Application Development Lifecycle

Stage 6 - Deployment and Maintenance

- **Objective:** Covering the final stages of the lifecycle, focusing on deployment strategies and ongoing maintenance.
- **Details:**
 - Containerization and orchestration tools for managing and scaling applications.
 - The importance of monitoring tools for performance tracking and identifying issues post-launch.
 - Establishing feedback loops with users to inform future development cycles.
 - ***AI: monitoring tools for detecting anomalies, logs crawlers, predictive analysis before visible real impact.***





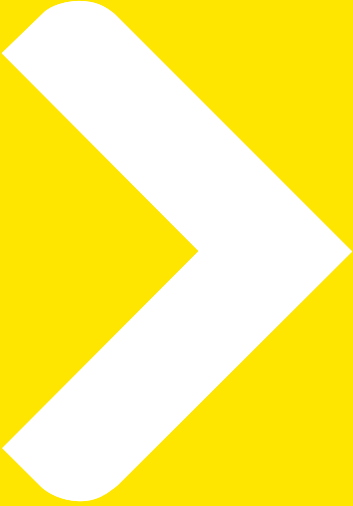
1. **Thorough Planning is Essential:** Success starts with clear goals, market research, and resource allocation.
2. **Requirements Drive Results:** Well-defined functional and non-functional requirements ensure the solution meets user needs.
3. **Design for Experience:** Prioritize user-centric design, accessibility, and responsiveness for broad adoption.
4. **Quality Implementation:** Adhere to coding standards, leverage automation (CI/CD), and foster collaborative development.
5. **Testing is Non-Negotiable:** Employ a mix of automated and manual testing to guarantee reliability and performance.
6. **Deployment & Maintenance Matter:** Use modern deployment strategies, monitor proactively, and maintain feedback loops for continuous improvement.
7. **Leverage AI:** Integrate AI tools throughout the lifecycle for smarter planning, design, coding, testing, and maintenance.



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Thank you
for your attention.

**Let's Make IT
happen**

